Safety Data Sheet: CHEM-AQUA 42171

Supercedes Date 08/13/2012 **Issuing Date** 12/28/2012

1. PRODUCT AND COMPANY IDENTIFICATION

Product Name CHEM-AQUA 42171
Recommended use Water treatment chemical Biocidal product
Information on Manufacturer
CHEM-AQUA, INC
BOX 152170
IRVING, TEXAS 75015

Product Code C743
Chemical nature Aqueous solution Alkaline
Emergency Telephone Number
CHEMTREC® 800-424-9300
Telephone inquiry
972-579-2477

2. HAZARD IDENTIFICATION

Color Yellow **Physical State** Liquid **Odor** Slight chlorine

GHS Classification

Physical Hazards

Substances/mixtures corrosive to metal	Category 1

Health Hazard

Skin Corrosion/Irritation	Category 1
Serious Eye Damage/Eye Irritation	Category 1

Other hazards
None

Labeling
Signal Word
DANGER



Hazard Statements
H314 - Causes severe skin burns and eye damage
H290 - May be corrosive to metals

Precautionary Statements
P280 - Wear protective gloves/protective clothing/eye protection/face protection
P260 - Do not breathe mists or vapors
P264 - Wash face, hands and any exposed skin thoroughly after handling.
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing Rinse skin with water/shower
P334 - Immerse in cool water/wrap with wet bandages
P363 - Wash contaminated clothing before reuse
P333 + P313 - If skin irritation or rash occurs: Get medical advice/attention
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P310 - Immediately call a POISON CENTER or doctor/physician
P304 + P340 - IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing.
P304 + P312 - IF INHALED: Call a POISON CENTER or doctor/physician if you feel unwell
P301+ P330 + P331 - IF SWALLOWED: Rinse mouth. Do NOT induce vomiting
P301+ P312 - IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell
P390 - Absorb spillage to prevent material damage
P406 - Store in corrosive resistant high density polyethylene (HDPE) container with a resistant inliner
P501 - Dispose of contents/container to an approved waste disposal plant.

11 % of the mixture consists of ingredient(s) of unknown toxicity

3. COMPOSITION /INFORMATION ON INGREDIENTS

Component	CAS-No	Weight %
Sulfamic acid, n-bromo, sodium salt	1004542-84-0	15-40
Sodium chloride	7647-14-5	7-13

Sodium hydroxide	1310-73-2	1-5

4. FIRST AID MEASURES

General advice	Do not get in eyes, on skin or on clothing. Do not breathe vapors or spray mist.
Eye Contact	Immediately flush with plenty of water. After initial flushing, remove any contact lenses and continue flushing for at least 15 minutes. Call a physician or poison control center immediately.
Skin Contact	Remove immediately all contaminated clothing. Wash off immediately with plenty of water for at least 15 minutes. Call a physician or poison control center immediately.
Inhalation	Move to fresh air. If not breathing, give artificial respiration. Call a physician or poison control center immediately.
Ingestion	Call a physician or Poison Control Center immediately. Do NOT induce vomiting. Never give anything by mouth to an unconscious person.
Notes to physician	The product causes burns of eyes, skin and mucous membranes. Control of circulatory system, shock therapy if needed.

5. FIRE-FIGHTING MEASURES

Flash Point Does not flash **Method** Not applicable
Flammability Limits in Air % Hydrogen, by reaction with metals. **Upper** 75 **Lower** 4
Suitable Extinguishing Media
Water spray. Foam. Alcohol-resistant foam. Carbon dioxide (CO2). Dry chemical. Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Specific hazards arising from the chemical
Contact with metals may evolve flammable hydrogen gas. Material can create slippery conditions.
Protective Equipment and Precautions for Firefighters
As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

NFPA	**Health** 3	**Flammability** 1	**Instability** 1
HMIS	**Health** 3	**Flammability** 1	**Instability** 1

6. ACCIDENTAL RELEASE MEASURES

Personal Precautions	Use personal protective equipment. Prevent further leakage or spillage if safe to do so. Material can create slippery conditions.
Environmental Precautions	Do not flush into surface water or sanitary sewer system.
Methods for Containment	Contain spillage, soak up with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and transfer to a container for disposal according to local / national regulations (see section 13).
Methods for Cleaning Up	Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust)
Neutralizing Agent	Acetic acid, diluted.

7. HANDLING AND STORAGE

Handling	Do not get in eyes, on skin or on clothing. Do not breathe vapors or spray mist.
Storage	Store in original container. Keep away from direct sunlight. Keep containers tightly closed in a dry, cool and well-ventilated place. Metal containers must be lined. Do not freeze.
Storage Temperature	**Minimum** 45 °F / 7 °C **Maximum** 100 °F / 38 °C
Storage Conditions	**Indoor** X **Outdoor** **Heated** **Refrigerated**

8. EXPOSURE CONTROLS / PERSONAL PROTECTION

Exposure Guidelines

Component	ACGIH TLV	OSHA PEL	NIOSH
Sulfamic acid, n-bromo, sodium salt	No data available	No data available	No data available
Sodium chloride	No data available	5 mg/m^3 PNOR (as solid)	No data available
Sodium hydroxide	Ceiling: 2 mg/m^3	TWA: 2 mg/m^3	IDLH: 10 mg/m^3 Ceiling: 2 mg/m^3

Engineering Measures	Use with local exhaust ventilation. Ensure adequate ventilation, especially in confined areas.
Personal Protective Equipment	
Eye/Face Protection	Tightly fitting safety goggles. Face-shield.
Skin Protection	Wear suitable protective clothing, Impervious gloves.
Respiratory Protection	In case of inadequate ventilation wear respiratory protection. When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.
General Hygiene Considerations	Remove and wash contaminated clothing before re-use.

9. PHYSICAL AND CHEMICAL PROPERTIES

Physical State	Liquid	**Viscosity**	Non viscous
Color	Yellow	**Odor**	Slight chlorine
Odor Threshold	Not applicable	**Appearance**	Transparent
pH	> 13	**Specific Gravity**	1.32
Evaporation Rate	0.38 (Butyl acetate=1)	**Percent Volatile (Volume)**	73.5
VOC Content (%)	0	**VOC Content (g/L)**	0
Vapor Pressure	11.8 mmHg @ 70°F	**Vapor Density**	0.6 (Air = 1.0)
Solubility	Completely soluble	**n-Octanol/Water Partition**	No data available
Melting Point/Range	No data available	**Decomposition Temperature**	No data available
Boiling Point/Range	No data available	**Flammability (solid, gas)**	No data available
Flash Point	Does not flash	**Method**	Not applicable
Autoignition Temperature	No information available.		
Flammability Limits in Air %	Hydrogen, by reaction with metals.	**Upper** 75 **Lower** 4	

10. STABILITY AND REACTIVITY

Chemical Stability	Stable. Hazardous polymerization does not occur.
Conditions to Avoid	Extremes of temperature and direct sunlight
Incompatible Products	Strong oxidizing agents, Acids, Metals.
Hazardous Decomposition Products	Carbon oxides, Nitrogen oxides (NOx), Sulfur oxides, Sulfur compounds, Sodium oxides, Bromine, Fumes of bromine, Hydrogen bromide.
Possibility of Hazardous Reactions	None under normal processing

11. TOXICOLOGICAL INFORMATION

Product Information

The following values are calculated based on chapter 3.1 of the GHS document (Rev. 3, 2009):

Oral LD50	No information available
Dermal LD50	No information available
Inhalation LC50	
Gas	No information available
Mist	No information available
Vapor	No information available

Principle Route of Exposure	Skin contact, Eye contact, Inhalation.
Primary Routes of Entry	None known
Acute Effects	
Eyes	Corrosive to the eyes and may cause severe damage including blindness.
Skin	Causes skin burns.
Inhalation	Harmful by inhalation. Causes burns.
Ingestion	Ingestion causes burns of the upper digestive and respiratory tracts. May be fatal if swallowed.
Chronic Toxicity	Inhaled corrosive substances can lead to a toxic edema of the lungs.
Target Organ Effects	Skin, Respiratory system.
Aggravated Medical Conditions	Skin disorders, Respiratory disorders.

Component Information

Acute Toxicity

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation	Draize Test	Other
Sulfamic acid, n-bromo, sodium salt	no data available	no data available	no data available	no data available	no data available
Sodium chloride	= 3 g/kg (Rat)	> 10 g/kg (Rabbit)	> 42 g/m^3 (Rat) 1 h	no data available	no data available
Sodium hydroxide	no data available	= 1350 mg/kg (Rabbit)	no data available	no data available	no data available

Chronic Toxicity

Component	Mutagenicity	Sensitization	Developmental Toxicity	Reproductive Toxicity	Target Organ Effects
Sulfamic acid, n-bromo, sodium salt	no data available	no data available	no data available	no data available	no data available
Sodium chloride	no data available	no data available	no data available	no data available	kidney
Sodium hydroxide	no data available	no data available	no data available	no data available	eyes, respiratory system, skin

Carcinogenicity — There are no known carcinogenic chemicals in this product.

Component	ACGIH	IARC	NTP	OSHA	Other
Sulfamic acid, n-bromo, sodium salt	not applicable	not applicable	not applicable	not applicable	not applicable
Sodium chloride	not applicable	not applicable	not applicable	not applicable	not applicable

Sodium hydroxide	not applicable	not applicable	not applicable	not applicable	not applicable

12. ECOLOGICAL INFORMATION

Product Information No information available.

Component Information

Component	Toxicity to Algae	Toxicity to Fish	Microtox	Water Flea	log Pow
Sulfamic acid, n-bromo, sodium salt	no data available	no data available	no data available	no data available	N/A
Sodium chloride	no data available	LC50 5560 - 6080 mg/L Lepomis macrochirus 96 h LC50 = 12946 mg/L Lepomis macrochirus 96 h LC50 6020 - 7070 mg/L Pimephales promelas 96 h LC50 = 7050 mg/L Pimephales promelas 96 h LC50 6420 - 6700 mg/L Pimephales promelas 96 h LC50 4747 - 7824 mg/L Oncorhynchus mykiss 96 h	no data available	EC50= 1000 mg/L 48 h EC50 340 7 - 469.2 mg/L 48 h	N/A
Sodium hydroxide	no data available	LC50 = 45.4 mg/L Oncorhynchus mykiss 96 h	no data available	no data available	N/A

Persistence and Degradability No information available.
Bioaccumulation No information available.
Mobility No information available.

13. DISPOSAL CONSIDERATIONS

Product Disposal Pesticide wastes are acutely hazardous. Improper disposal of excess pesticide or rinsate is a violation of federal law. If these wastes cannot be disposed of by use according to label instructions, contact your state pesticide or environmental control agency.

Container Disposal Empty containers should be taken for local recycling, recovery, or waste disposal.

14. TRANSPORT INFORMATION

DOT

- **Proper Shipping Name** Corrosive liquid, n.o.s.
- **Hazard Class** 8
- **UN-No** UN1760
- **Packing Group** III
- **Description** UN1760, Corrosive liquid, n.o.s.,(Bromide Salts), 8, PG III

TDG

- **Hazard Class** 8
- **UN-No** UN1760
- **Packing Group** III

ICAO

- **UN-No** UN1760
- **Proper Shipping Name** Corrosive liquid, n.o.s.
- **Hazard Class** 8
- **Packing Group** III
- **Shipping Description** UN1760, Corrosive liquid, n.o.s.,(Bromide Salts), 8, PG III

IATA

- **UN-No** UN1760
- **Proper Shipping Name** Corrosive liquid, n.o.s.
- **Hazard Class** 8
- **Packing Group** III
- **ERG Code** 8L
- **Shipping Description** UN1760, Corrosive liquid, n.o.s (Bromide Salts), 8, PG III

IMDG/IMO

- **Proper Shipping Name** Corrosive liquid, n.o.s.
- **Hazard Class** 8
- **UN-No** UN1760
- **Packing Group** III

EmS No.	F-A, S-B
Shipping Description	UN1760, Corrosive liquid, n.o.s.(Bromide Salts), 8, PG III

15. REGULATORY INFORMATION

Inventories

TSCA	Complies
DSL	Does not Comply

U.S. Federal Regulations

SARA 313

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and and Title 40 of the Code of Federal Regulations, Part 372.

SARA 311/312 Hazardous Categorization

Acute Health Hazard	Chronic Health Hazard	Fire Hazard	Sudden Release of Pressure Hazard	Reactive Hazard
Yes	No	No	No	No

CERCLA

Component	Hazardous Substances RQs	CERCLA EHS RQs
Sulfamic acid, n-bromo, sodium salt	Not applicable	Not applicable
Sodium chloride	Not applicable	Not applicable
Sodium hydroxide	1000 lb	Not applicable

Canada

This product may not be commercially placed on the market in Canada.

WHMIS Hazard Class

Not applicable

16. OTHER INFORMATION

Prepared By	Rachael Mohochi
Supercedes Date	08/13/2012
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Reason for Revision	No information available.
Glossary	No information available.
List of References.	No information available.

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